

Assignment 6 — Convolutions in Neural Networks

Problem 1: Manual 1D Convolution

Given input $\vec{x} = [3, 1, 4, 1, 5, 9, 2, 6]$ and kernel $\vec{w} = [1, 0, -1]$ with stride 1 and valid padding:

- (a) How many output elements will this convolution produce? Show your calculation using the output size formula.
- (b) Compute all output values by hand.
- (c) What kind of local pattern does this kernel detect? Explain by examining the structure of the weights.
- (d) At which output index is the absolute value largest? What does this tell you about the input at that location?

Problem 2: Stride and Padding Effects

Consider input $\vec{x} = [2, 5, 3, 8, 1, 7, 4, 6, 9, 0]$ (size $n = 10$) and kernel $\vec{w} = [\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ (size $k = 4$).

- (a) Compute the output size for each combination:

	Valid padding ($p = 0$)	Zero padding ($p = 2$)
Stride $s = 1$	?	?
Stride $s = 2$	?	?
Stride $s = 3$	?	?

- (b) For stride $s = 2$, valid padding, compute the first three output values.
- (c) Describe the physical effect of this kernel on the input signal. Why might this be useful in a noisy environment?

Problem 3: 1D Convolution as Matrix Multiplication

For input $\vec{x} = [x_0, x_1, x_2, x_3]$ and kernel $\vec{w} = [w_0, w_1]$ with stride 1 and valid padding:

- (a) Write the output vector $\vec{y}$ as a matrix-vector product $\vec{y} = W\vec{x}$. Explicitly write out the weight matrix W .
- (b) What are the dimensions of W ?

(c) Verify that W is sparse. What fraction of entries in W are zero?

(d) Now write the weight matrix for the same kernel but with stride 2. What changes?

Problem 4: Manual 2D Convolution

Given the 4×4 input image and 2×2 kernel:

$$X = \begin{bmatrix} 10 & 20 & 30 & 40 \\ 50 & 60 & 70 & 80 \\ 90 & 100 & 110 & 120 \\ 130 & 140 & 150 & 160 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

(a) What is the output size with stride $[1, 1]$ and valid padding?

(b) Compute the full output matrix.

(c) Interpret the kernel: what local pattern does $W = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ detect? Where in the output do you see the strongest response, and why?

Problem 5: Image Smoothing vs. Edge Detection

Consider a 5×5 grayscale image:

$$X = \begin{bmatrix} 100 & 100 & 100 & 100 & 100 \\ 100 & 100 & 100 & 100 & 100 \\ 10 & 10 & 10 & 100 & 100 \\ 10 & 10 & 10 & 100 & 100 \\ 10 & 10 & 10 & 10 & 10 \end{bmatrix}$$

(a) Apply the smoothing kernel $W_s = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ with stride $[1, 1]$, valid padding. Compute the output at positions $(0, 0)$, $(1, 1)$, and $(2, 2)$.

(b) Apply the vertical edge detection kernel $W_e = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$ with stride $[1, 1]$, valid padding. Compute the output at positions $(1, 1)$, $(1, 2)$, and $(2, 1)$.

(c) Which kernel produces larger responses at the boundary between the bright (100) and dark (10) regions? Explain why.

Problem 6: Output Size in Multiple Dimensions

A color image has dimensions $[H, W] = [224, 224]$ with $C = 3$ input channels.

- (a) A Conv2d layer has kernel size $[3, 3]$, stride $[1, 1]$, and padding $p = 1$. What is the spatial output size?
- (b) A second Conv2d layer has kernel size $[5, 5]$, stride $[2, 2]$, and valid padding ($p = 0$). Applied to the output of (a), what is the new spatial output size?
- (c) A third Conv2d layer has kernel size $[3, 3]$, stride $[2, 2]$, padding $p = 1$. Applied to the output of (b), what is the new spatial output size?
- (d) If each layer has 64 output channels, what is the total number of parameters (weights only, no bias) in the three layers combined? Assume 3 input channels for layer 1, and 64 for layers 2 and 3.

Problem 7: 3D Convolution for Motion Detection

A video has 8 grayscale frames of size 16×16 . We apply a 3D convolution with:

- Kernel size $[k_T, k_H, k_W] = [2, 3, 3]$
- Stride $[1, 1, 1]$
- Valid padding

- (a) What is the output size $[o_T, o_H, o_W]$?
- (b) A motion detection kernel has $W_{t=0} = -\mathbf{1}_{3 \times 3}$ and $W_{t=1} = +\mathbf{1}_{3 \times 3}$ (where $\mathbf{1}_{3 \times 3}$ is a 3×3 matrix of ones). At a particular slide stop, the 3×3 spatial patch in frame t has all values 50 and in frame $t + 1$ all values 50 (no motion). What is the output?
- (c) At another slide stop, the 3×3 patch in frame t has all values 50 and in frame $t + 1$ all values 200. What is the output? What does this indicate?

Problem 8: Transposed Convolution

A 1D convolution has kernel $\vec{w} = [w_0, w_1, w_2]$, input size $n = 5$, stride 1, valid padding.

- (a) Write the 3×5 weight matrix W for the forward convolution.
- (b) Write the 5×3 transposed weight matrix W^T .
- (c) Given output $\vec{y} = [2, -1, 3]$, compute $\tilde{x} = W^T \vec{y}$ (in terms of w_0, w_1, w_2).
- (d) Is $\tilde{x} = \vec{x}$ (the original input)? Explain why or why not.
- (e) Compute the output size of a transposed convolution with input size $n' = 3$, stride $s = 2$, kernel size $k = 3$, and padding $p = 0$.

Problem 9: Max Pooling vs. Average Pooling

Given the 6×6 feature map:

$$F = \begin{bmatrix} 1 & 3 & 2 & 8 & 0 & 5 \\ 7 & 4 & 6 & 1 & 3 & 2 \\ 0 & 9 & 5 & 4 & 7 & 1 \\ 3 & 2 & 8 & 6 & 0 & 9 \\ 5 & 1 & 0 & 3 & 8 & 4 \\ 6 & 7 & 2 & 1 & 5 & 3 \end{bmatrix}$$

- (a) Apply 2×2 max pooling with stride 2. Write the full 3×3 output.
 - (b) Apply 2×2 average pooling with stride 2. Write the full 3×3 output.
 - (c) Apply 3×3 max pooling with stride 3. Write the output and state its size.
 - (d) Explain why max pooling is generally preferred over average pooling in classification CNNs.
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Problem 10: CNN Architecture Design

You are designing a CNN for classifying 28×28 grayscale images into 10 classes.

- (a) Fill in the output dimensions for this architecture:

Layer	Configuration	Output Size
Input	—	$1 \times 28 \times 28$
Conv2d	16 filters, 3×3 , stride 1, pad 1	?
ReLU	—	?
MaxPool2d	2×2 , stride 2	?
Conv2d	32 filters, 3×3 , stride 1, pad 1	?
ReLU	—	?
MaxPool2d	2×2 , stride 2	?
Flatten	—	?
Linear	10 outputs	?

- (b) How many learnable parameters (weights + biases) does each Conv2d layer have?
- (c) How many parameters does the final Linear layer have?

(d) Compare the total parameter count to a single fully connected layer that maps the $28 \times 28 = 784$ input directly to 10 classes. What is the ratio?